Final Business Analyst Report

Customer Churn Analysis for a Telecommunications Company

Segmentation, Artificial Neural Networks and Evidence-Based Retention Planning

Project	Customer Churn Prediction and Segmentation
Document	Final analytical report
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Evidence base: 6,741 processed customer records, four published customer clusters and three independently evaluated classification models.

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The analytical results in this report describe the retrained project and the supplied 6,741-row datasets. Earlier figures reported in the sample document are not reused.

1. Executive Summary

This report appraises the work on a project that has been developed to study customer churn in the telecommunications sector from cleaned and clustered data provided in the Stage 2 repository as a reproducible project. 6,741 customers of which 1,791 are churn customers are included in the analysis. This represents a churn proportion of 26.57%. The project offers a descriptive approach that is followed by individual predictions, and compares the artificial neural network with the logistic regression and random forest classifiers. Unlike an earlier exploratory notebook, any model building and threshold decisions are kept in a separate validation group. The cluster assignments which have been published show remarkable 'clustering around' loss in Cluster 3. It has 2,146 customers with a churn rate of 48.84%, compared with 4.80% among the 1,124 customers in Cluster 1. The term "group identities" refers to the identities provided by the clustering file and is not the same as the newly created cluster identifiers from training. Tenures and charges are already normalized in source data and values reported will be as scores and not as actual months, or currencies. This distinction affects the interpretation and a possible business intervention. We chose the validation average precision method to select the ANN. It achieved ROC-AUC, PR-AUC, and Churn recall of 0.810, 0.587, and 91.90%, respectively, on the untouched test set. At the decision threshold corresponding to this recall, however, precision was 39.98%, with an accuracy of 61.23%. On the test set, logistic regression outperformed the ANN on ROC-AUC and PR-AUC. The ideal business approach, therefore, is to test all risk-ranked contact lists and measure retention success, rather than assume the ANN is a safe bet. The results should be given as a compromise (not as one absolute percentage). A threshold that captures almost everyone who churns will also capture extra contacts with customers who do not churn. The correct ratio will depend on the range of operational capability, the value to the customer, and the cost of the proposed assistance. None of the above cost variables is available in the information, so a profitable threshold cannot be determined.

2. Project Background and Business Problem

A telecom provider's subscription revenue exists only when customers are active. Cancellations can adversely affect future revenue and may require marketing spend to replace the customer. The question of whether or not a customer will leave is practical. Whether an intervention can be provided at a sensible cost and whether it changes the outcome. Only a churn probability makes sense as part of this broader retention decision. The project draws on the public Deliverables repository for Stage 2, maintained by Goreja (n.d.). It provides cleaned data and a clustered customer extract, and the rebuilt analysis offers a more structured comparison of models. Of the 6741 records, almost a quarter of customers reported that they had already left. More significantly the risk is not uniformly distributed. It focuses on the epicentres of losses, the certainty of churn detection, and management's initiation of selective customer support in a pilot program, without discounting all manner of customers. A standard precision metric alone would not suffice to answer these questions. The overwhelming majority of the customers who stayed (73.43%) would make a stay-only prediction seem correct, even though right predictions of cancelation would not be granted. The business problem thus requires steps to assess ranking quality and estimate the false-contact burden for the selected action threshold.

3. Project Objectives and Scope

The overall aim is to convert prepared customer information into a defensible churn-risk assessment and a practical basis for retention planning. The analytical objectives cover both the structure of the customer population and the reliability of individual predictions.

- Confirm the integrity and meaning of the two supplied customer datasets.
- Identify meaningful differences in churn rates across the four published customer clusters.
- Develop comparable ANN, logistic regression and random forest classifiers.
- Use validation data to select models and thresholds without using test outcomes for tuning.
- Translate tested predictive results into retention priorities, controls and measurable indicators.

The scope includes exploratory analysis, K-Means clustering fitted on training records, supervised modelling, graphical evaluation, saved inference artefacts and recommendations. It does not include live customer contact,

confirmed financial returns, causal estimation of churn drivers, or deployment within an operational billing platform.

The reported work is consequently an offline validation study. Its stakeholder value comes from identifying testable decisions and explaining uncertainty, rather than presenting the project as an established commercial retention system.

4. Business Analyst Role and Contribution

The business analyst contributes by translating several technical outputs into a single decision framework. This includes verifying that the reported model performance reflects the correct data version, investigating false alerts and missed churner costs, and summarising the shortcomings of the variables available to the model. Additional precautions apply here, as the prior sample report had different cluster risk rankings and a different row count. The historical accounts do not apply to the reconstructed project. The resulting workflow is delivered as a notebook for Colab (or other), a training script that can be reused, test cases, utils for prediction, model files, score tables and charts. The analyst then has to determine what they've found that might help a low risk pilot, and what claims need to be investigated further. Probabilities that are "reported" do not mean that a specific incentive ensures that cancellation is avoided, but rather show some statistical association. Before crediting any saved revenues for the proposed programme, a comparison group is required.

5. Dataset and Data Preparation Summary

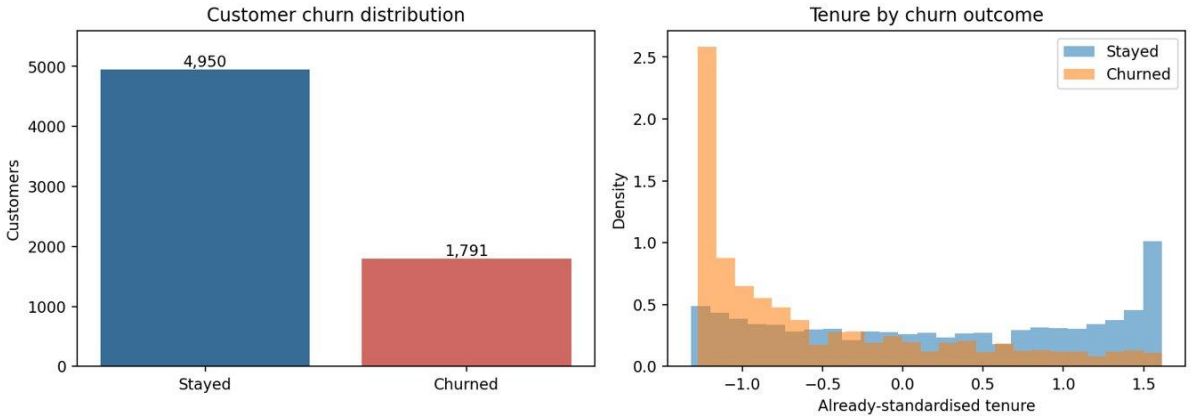
The modelling data source has 6741 rows and 13 columns (including binary target). There are 12 predictor columns. The clustered reference includes Cluster, PCA1, and PCA2, with the same observation order across rows. 4950 customers were retained, and 1791 churned. This imbalance matters because a single model that only predicts retention can appear highly accurate without flagging real churn.

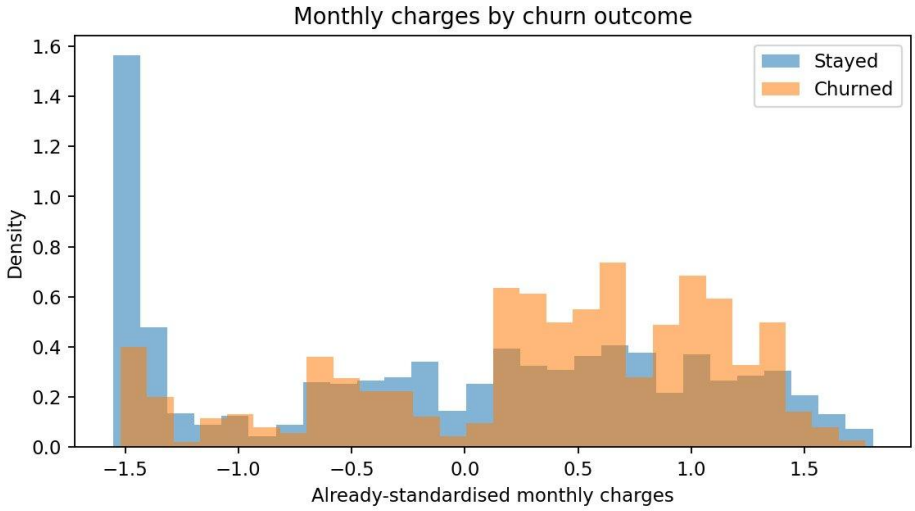
Table 1. Dataset structure and analytical purpose

Dataset element	Observed value	Analytical role
Customer records	6,741	One row per customer
Predictor fields	12	Encoded characteristics, contracts and service attributes
Non-churn / churn	4,950 / 1,791	Target balance for binary classification
Published clusters	4	Descriptive segmentation reference
PCA values	PCA1 and PCA2	Visualisation only, excluded from predictors

5.1 Data preparation actions

The source numerically represents gender, service selections, contract category, and other binary variables. Tenure and MonthlyCharges were already standardised upstream. Predictor order and missing values and separation of target were verified during the rebuilding process. The proportion of churn in each model-development group was roughly equalized via stratification. We used only training data for model preprocessing and to generate new K-Means centres, without retraining on validation/training data. There is a historical limitation yet to be resolved. The source does not include the raw billing numbers or the scaler used to standardise tenure and charges. This means the upstream scaling process cannot be audited, and raw-currency prediction is not possible from the saved pipeline without rescinding that scaling. Consistent fit and transform procedures are applied in the rebuilt workflow, consistent with the modelling procedures used in scikit-learn, the code accompanying which was well documented by Pedregosa et al. (2011).





6. Methodology Overview

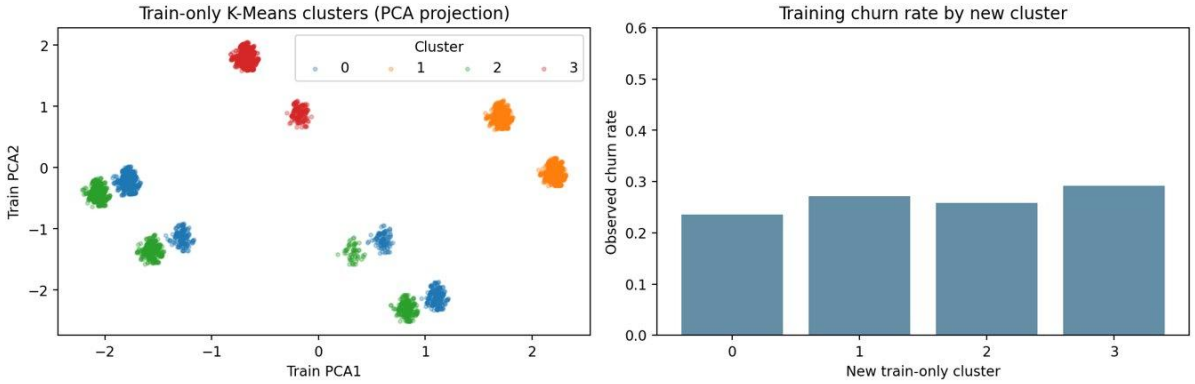
Design, validation and testing of the work is based on a train. The records are divided into a 64% training set, a 16% validation set, and a 20% untouched test set. The resulting sizes are 4,313, 1,079 and 1,349 customers. Four fresh customer segments are created using K-Means with training-only fitting. One-hot indicators for these new clusters are then added to the 12 original customer variables and consumed by the three prediction models.

Table 2. Modelling stages and safeguards

Stage	Method	Output
Training	4,313 customers; transformations fitted	Learnt model weights and cluster centres
Validation	1,079 customers; threshold and selection	Selected ANN and risk threshold
Independent testing	1,349 customers; no refitting	Comparable final performance scores
Inference	Saved model and preprocessing objects	Repeatable churn probability estimates

6.1 Why clustering and predictive modelling were both useful

Clustering is used for taking decisions about clusters of customers with similar encoded customer characteristics. Prediction involves a different question and predicts the risk of an individual record. The published clusters are used to interpret the customer segments, and the newly fitted clusters are used to predict the features. Separation prevents, for example, recycling cluster assignments based on the full data set into a supposedly ‘unseen’ test evaluation. The validation choice is also meant to be independent of the final assessment group, as discussed by scikit-learn developers (n.d.).



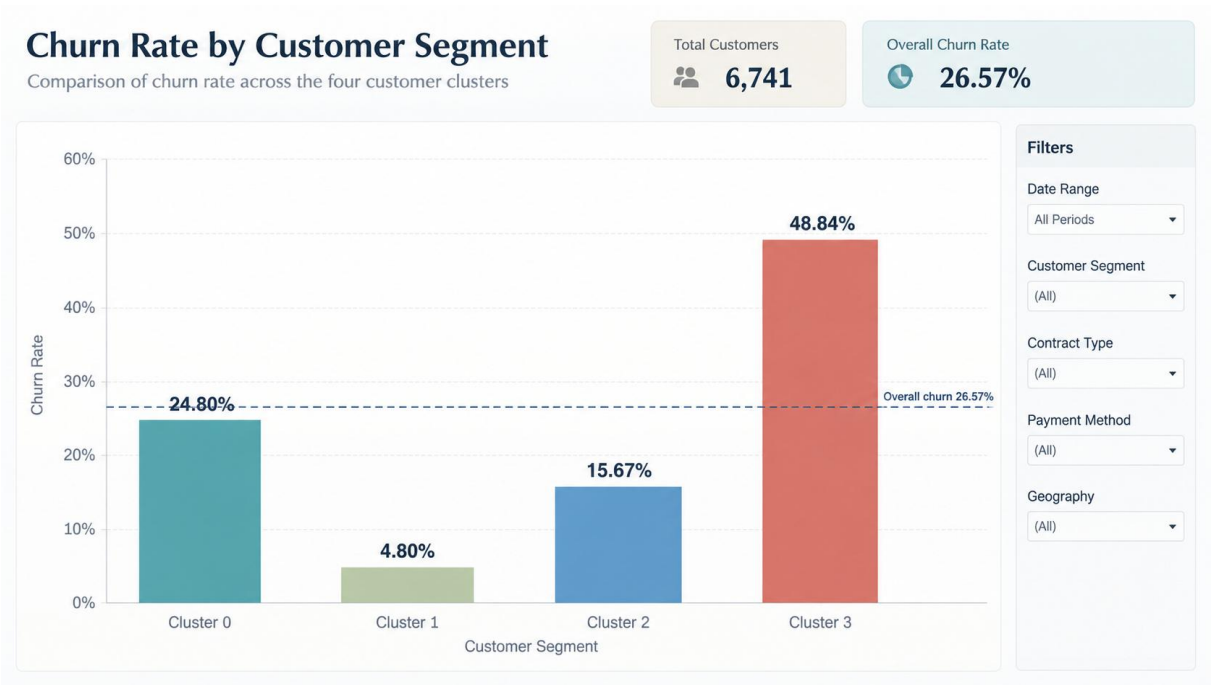
7. Detailed Clustering Analysis Findings

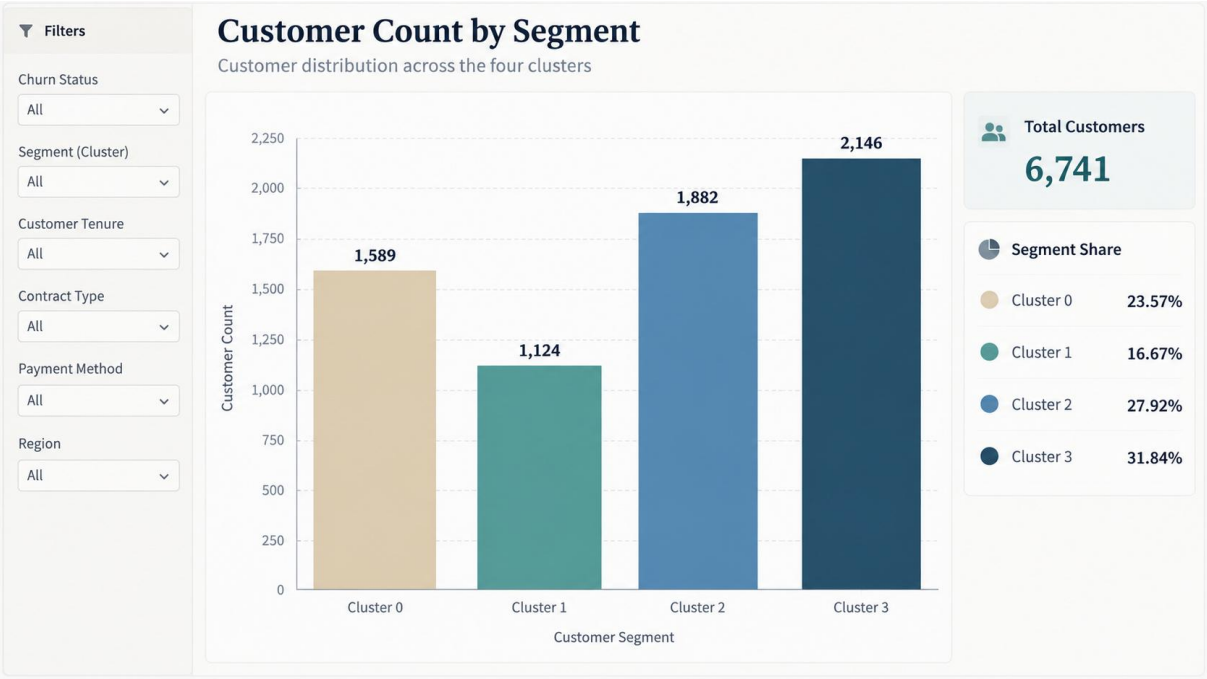
There are four published clusters in the source repository. Much of the profile information differs from the labels and values in the previous sample report. Based on the data shown, Cluster 3, not Cluster 2, has the highest churn rate. The charts illustrate only descriptive relationships of association and are not a verification of segment identities. Original labels should therefore be treated as numerical labels until the company defines them and explains how they should be interpreted within its business. The size of each segment can also influence where we set intervention points. Cluster 3 has nearly 30% of all the observations and has a significantly higher churn percentage than any other published cluster. Cluster 1's strong performance is not driven by a single characteristic, and it is a good comparison for low risk. Features such as alternative contract mixes, service selection, and combinations of demographics may drive the cluster figures. Before assigning behavioural names, further investigation should be done using cross-tabulations and customer interviews.

Table 3. Published customer segments

Segment	Customers	Churn	Senior share	Mean tenure*	Mean charges*
Cluster 0	1,589	24.80%	7.24%	-0.86	-1.03
Cluster 1	1,124	4.80%	7.56%	0.90	-0.99
Cluster 2	1,882	15.67%	21.68%	1.07	0.95
Cluster 3	2,146	48.84%	24.32%	-0.70	0.56

*Tenure and charges are standardised values rather than months or currency amounts.





7.1 Cluster 0: Shorter Tenure and Lower Charges

Cluster 0 has 1,589 customers and a churn rate of 24.80%, slightly lower than the overall churn rate. The group has shorter-tenured standardised contracts, and decidedly lower Charge scores. These variables alone are not sufficient to infer low income or confirmed price sensitivity. Some plan comparisons and support for service onboarding would have been useful in a pilot. Before creating discounts that may be unnecessary, the company should check for and track early account friction.

7.2 Cluster 1: Longer Tenure and Lower Charges

Cluster 1 has the lowest churn (4.80%) and a total of 1,124 customers. It has a higher tenure score and a lower charge score, suggesting its relationship in the published segmentation is fairly stable. This is an important sample set that might be used to determine if there is a correlation between a relationship duration and decreased exposure to the cost of continuing service. Avoid spending money on incentives for customers likely to remain in service anyway, and instead focus on keeping services running and preventing unnecessary failures.

7.3 Cluster 2: Longer Tenure and Higher Charges

Although standardised charge levels indicate relatively low churn rates, this is not the case for Cluster 2, which has 1,882 customers and 15.67% churn. The ratio of elderly persons is not 21.68% but rather includes both elderly and youth. This could be partly due to a longer tenure. So the obvious answer is to look at how satisfied customers are with the premium service and their tendency to renew. We cannot conclude that churn is driven by higher charges in this segment, as it is less likely to churn than the overall sample.

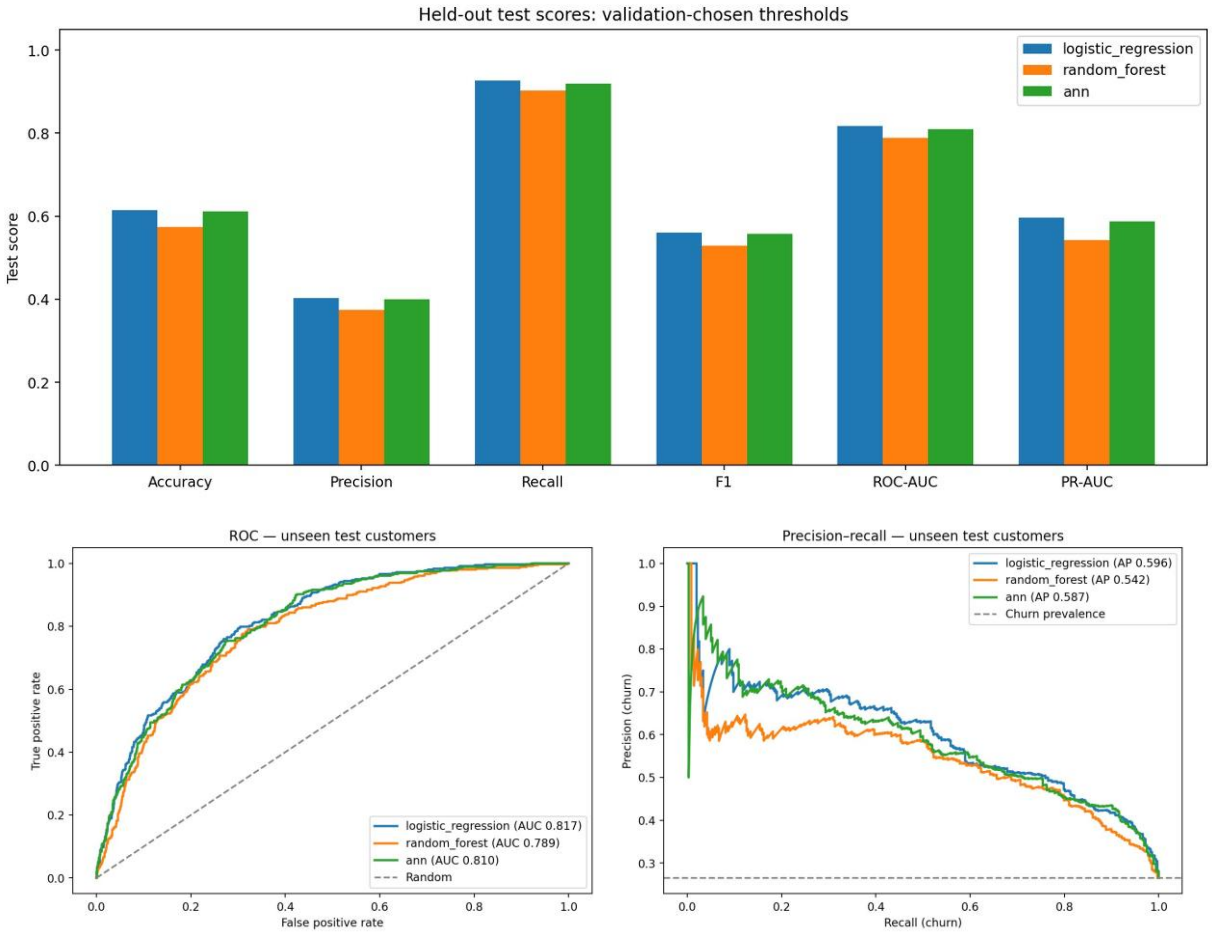
8. Detailed ANN Predictive Modelling Findings

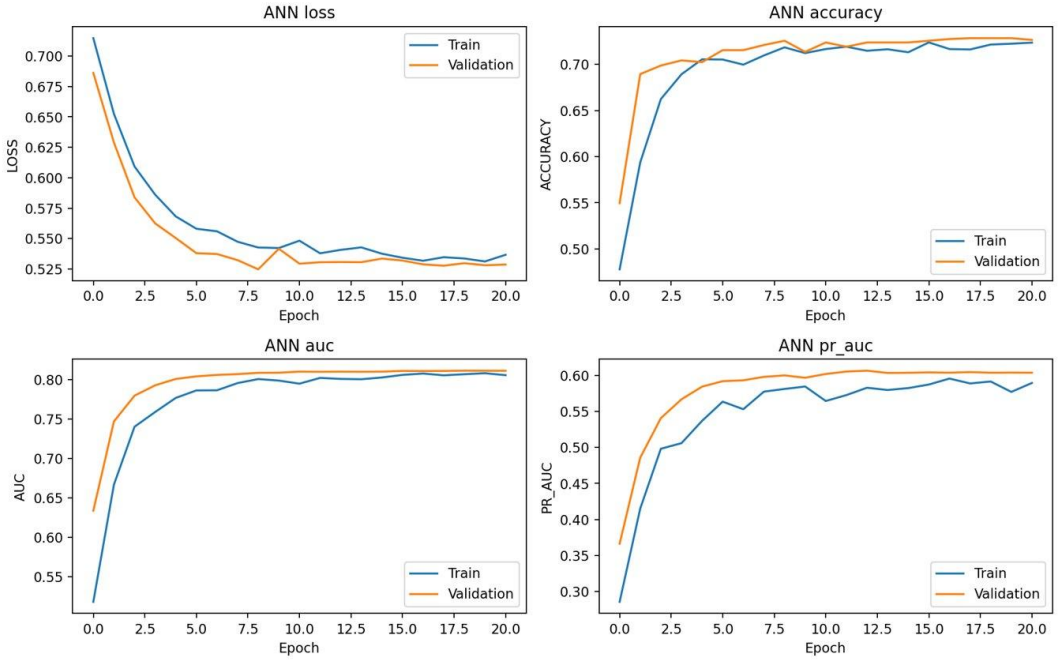
Keras was used to retrain a feed-forward artificial neural network with a probabilistic churn output and binary cross-entropy loss. When constructing the model, sixteen final predictors were used, consisting of twelve customer variables and four new cluster indicators. Training was requested, a maximum of 45 epochs was desired and training stopped after 21 epochs. The workflow was also used with logistic regression and random forest with the same held-out observations. Decision thresholds were determined by using validation F2, which puts more emphasis on getting the actual churners right, while validation average precision was used to select the models. Keras Team (n.d.) explains the training/evaluation interfaces that can be used in such networks, and Kingma and Ba (2015) give the original formulation of the Adam optimisation method. The ANN's superiority was also evident in its slightly higher validation PR-AUC (0.6067) than logistic regression (0.6043). On the held-out test set, however, logistic regression outperformed on ROC-AUC, PR-AUC and F1. This does not cancel out the

original validations made. It indicates that the differences are insignificant, and stakeholders should not presume a neural network will perform better than the independent test evidence. Saito and Rehmsmeier (2015) explain why precision-recall analysis's useful when, as is often the case, a positive outcome is relatively rare. The test results should be considered predictors based on a single test set of 1,349 customers. If the areas under the curve differ by a few thousandths, they are not sufficiently significant to draw strong conclusions on the absence of repeat evaluation and confidence intervals. Logistic regression is a simpler model that may be considered next because it achieves slightly better results and is easier to examine and maintain. More testing might determine whether the new customer segments still offer a small edge.

Table 4. Held-out model results at validation-selected thresholds

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
ANN	0.612	0.400	0.919	0.557	0.810	0.587
Logistic regression	0.615	0.402	0.927	0.561	0.817	0.596
Random forest	0.575	0.375	0.902	0.530	0.789	0.542



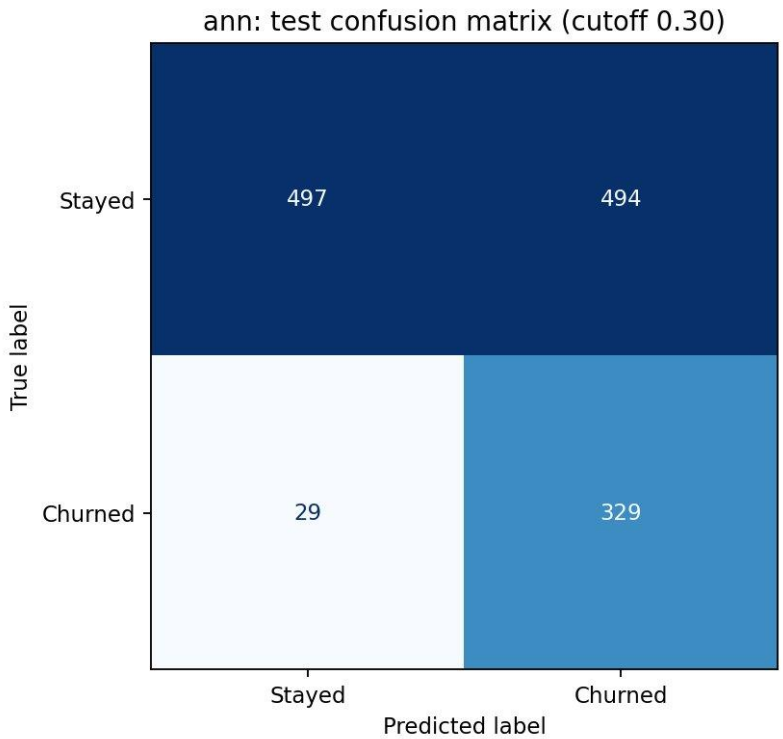


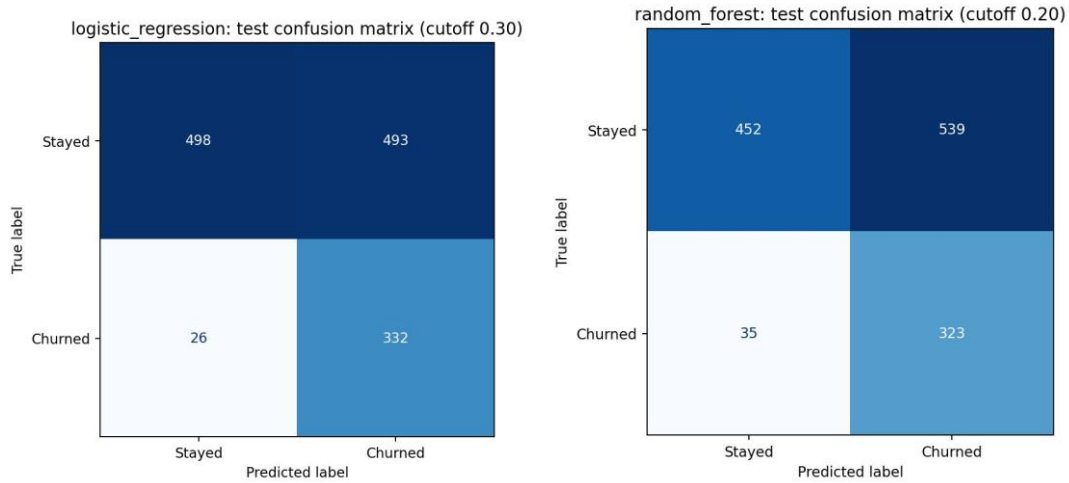
8.1 Confusion matrix interpretation

The results for the ANN test prediction showed 329 out of 358 actual churners are predicted accurately and 29 contemporary customers were overlooked. Of those who remained (991 in all), 497 were recognized correctly and 494 were unnecessarily marked. This yields high recall but also creates a significant operational workload. A detection threshold of 0.298 was purposefully chosen to be in favor of detection. At the ordinary level (0.50), ANN accuracy rises to 71.98%, and recall drops to 75.70%. It is to be hoped that both operating points will be available for use when shopping the customer contacts.

Table 5. ANN confusion matrix on the test set

Actual outcome	Predicted stay	Predicted churn
Stayed	497	494
Churned	29	329





9. Combined Factors Contributing to Churn

The analysis focuses on the profiles of customer circumstances and not causal drivers. The clusters with the highest churn rate were those with shorter tenure and higher standardised charges. Customer age also varies, but in Cluster 3, 24.32% of customers are senior citizens, and most are not. Overall, the senior group experiences churn rates around 41.24%, while the non-senior group experiences churn rates around 23.61%. Comparisons are descriptive, and may be the result of other differences in plan type or accessibility of services.

Table 6. Signals, evidence and interpretation

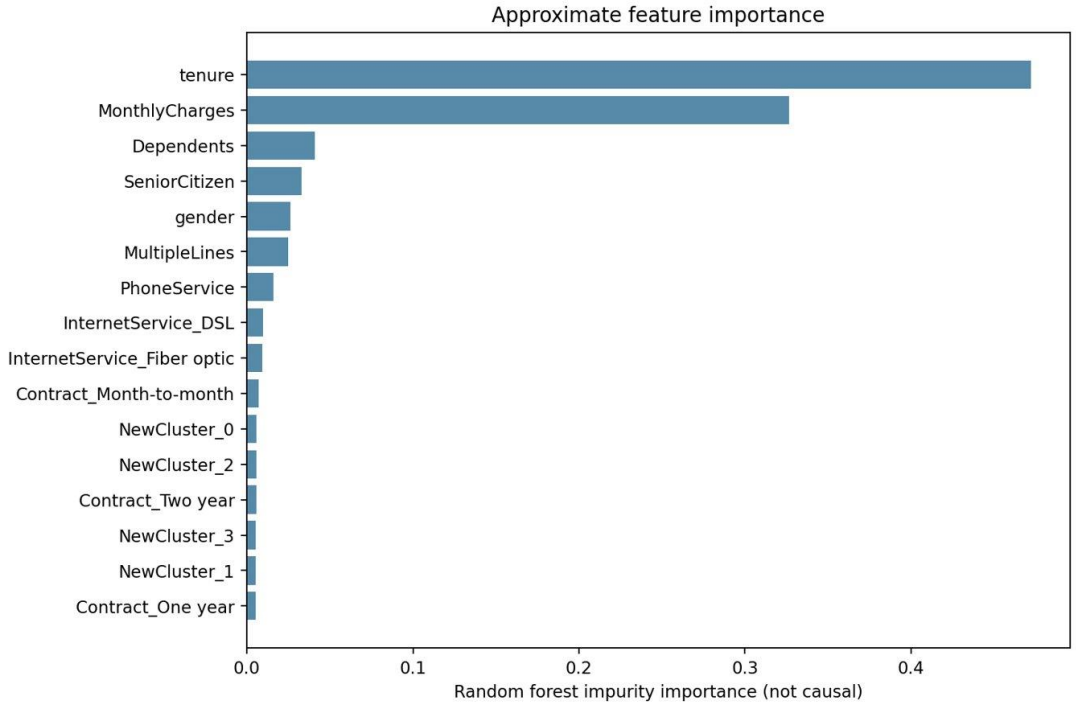
Signal	Evidence from supplied data	Caution
Customer tenure	Cluster 3 mean -0.70; churn 48.84%	Association is not proof of an onboarding problem
Charge exposure	Cluster 3 mean 0.56; Cluster 2 mean 0.95	Higher charges alone cannot explain segment risk
Senior status	Senior churn 41.24%; non-senior 23.61%	Other customer characteristics may confound difference
Predictive probability	ANN detects 329 of 358 churners	A flagged customer may remain without intervention

7.4 Cluster 3: Shorter Tenure and Higher Charges

The 2,146 customers that exist within Cluster 3 can be seen to represent the most defined churning, with 48.84% of the customers identified for study within this cluster. It has above-average standardised charges and is the only one of the four with shorter tenure and a higher proportion of senior customers (24.32%). Establishing of an early relationship management is the defensible operational priority. This segment of customers should be provided with more transparent onboarding process, swift solutions to the customer's problems, and transparent plan reviews. The effectiveness of any intervention cannot be based on cluster membership alone, and needs to be tested. It is important to understand the magnitude of this concern, and a simple prioritisation calculation can help. Around 1048 customers from Cluster 3 are marked as churned in the churn extract provided; this constitutes more than half of the churn in the extract. This concentration provides a reasonable basis for the initial pilot. However, it does not indicate which customers could have been retained, and the anticipated retention rate is unknown.

9.1 Main business interpretation

The main conclusion is that there is no single factor which can be said to be the appropriate response. Note that segments correlate with the identification of risk concentrations, and predictive scores allow an individual to be ranked within those segments. Use the model to assess customer experience (and service needs), and make offers based on controlled business experiments, rather than automatic discounting.



10. Retention Strategy Recommendations

Any targeted retention programme should begin by identifying customers at a higher-than-normal risk level, then assess the cluster and the cost of the appropriate response. The selected ANN threshold is likely to result in many non-churners being marked, therefore, large discounts must not be made automatically. Low-cost contacts, like service checks or bill clarifications, may be offered generously in exchange for more data on disengagement; higher-cost contacts may require more data on disengagement. The study must include a test of an untreated or business-as-usual (BAU) comparison group.

Table 7. Proposed actions by observed group

Customer group	Proposed intervention	Monitoring focus
Cluster 3	Early-tenure support, plan clarity and proactive problem resolution	Churn and contact cost per retained customer
Cluster 0	Onboarding review and easy-to-understand plan comparisons	Offer uptake and retention without deep discounting
Cluster 2	Premium-service quality and renewal reviews	Complaint reduction and contract continuation
Cluster 1	Reliable service and loyalty maintenance	Unexpected changes in low baseline churn
High ANN risk	Ranked contact queue subject to capacity limits	Incremental retention against a control group

10.1 Recommended retention campaign design

The trial should start with a precise customer list and a set contact capacity. Customers may be sorted by ANN risk, and the Publication of the Cluster Label provides a descriptive outlook on the contact queue. Offer the ordinary service to a comparable group served by randomisation to estimate its impact. The performance rates should be provided regarding contact success, retention, customer complaints and cost. This doesn't imply that a high prediction score should be used to indicate a successful retention campaign.

10.2 Why Cluster 3 should be priority

Cluster 3 is the biggest published portion and also the highest churn cluster observed. It is a logical area to test improvements in early relationship management due to the risk concentration in this area. However, campaign return depends on customers responding to the selected action. The top performing segment for a pilot isn't always the segment that will offer the highest financial benefit and a controlled study is required to so.

11. Implementation Roadmap and Stakeholder KPIs

Incentives should be phased to let the business learn before major incentives. The first month should confirm eligibility, size up the program, secure needed operational approvals, and build a contact list. The second month must be an ongoing, controlled pilot, with consistent results recorded. The third month will test the added impact by changing the decision rule. Start model monitoring as soon as scores are available, for use in business.

Table 8. Implementation timetable

Period	Operational activity	Measure of completion
Days 1–30	Define pilot, contacts, comparison group and consent controls	Approved protocol and balanced groups
Days 31–60	Run limited Cluster 3 retention contact programme	Delivery, uptake and service response recorded
Days 61–90	Compare churn outcomes and intervention cost	Incremental retention and net benefit review
Ongoing	Review drift, fairness, thresholds and retraining needs	Monthly governance record

11.1 Suggested KPIs

The core indicators are the ones that should be included as the observed churn by published cluster, the true churners (those that are detected per 100 customers contacted) and also the unnecessary contacts, campaign acceptance, average cost of the contact, incremental retention and also net amount of revenue that is protected. All metrics (precision, recall, F1, ROC-AUC and PR-AUC) should be kept in the model reporting. Even if more churners are found, the threshold should not be set solely on that basis, as the costs of such contacts may outweigh the benefits. To achieve this, threshold reviews require an agreed calculation of service capacity and business value, following Fawcett's evaluation framework described above.

12. Project Limitations

The most important missing information in the dataset is the original (untransformed) billing and tenure data. Standardised numbers are already supplied with cleaned file and the meaning of the numbers cannot be reconstructed in the training pipeline. Information used in the historical fitting process may have come from outside a later train split. The published cluster and PCA columns were also generated on the entire pre-processed dataset, which is why the column is not included in the predictive workflow as a model input. These new training-only clusters reduce one source of leakage, but can't correct past training that includes preprocessing. Data is a snapshot in time and has a limited range of customer attributes. Customer complaints, customer outages, payment delays, real costs, and real-time events on customers' accounts are not available. The model comparison uses one defined split and does not involve repeated external validation. There are many false positives at a high-recall operating point, and the probabilities are not independently calibrated for financial decision-making. The ANN was trained using a PyTorch backend running on Colab, compatible with Keras, and the backend execution for this scope has been provided. Still, its compatibility with Colab production has not been directly verified. The results too, are limited to the set of records. No later cut of the customer cohorts is given for temporal validation and no information was given regarding when each churn label was known in relation to the predictors. Therefore, the reporting cannot ensure the lead time a contact centre would need to act before cancellation. The absence of these timestamps is a fundamental limitation of the possible deployments. The dashboards you are provided are in Tableau format and are only images of the dashboards, not a Tableau workbook. They have a visible filter, but it is not interactive. No inferences are made from the filter labels for processed data in fields like region, payment method and date range. However, the cluster counts and churn rates displayed do correspond to the provided clustering data.

13. Future Improvements

The initial technical enhancement is to retrieve all raw source data and embed all transformations within a fully replicable cross-validation pipeline. Repeated Stratified Evaluation or nested selection would provide more

reliable uncertainty estimates than a single split. Calibration plots may help determine if there is a relationship between the claimed risk (30% churn) and observed risk. Then, thresholds could be set based on contact capacity, discount value, and retention gain, according to cost-sensitive optimisation. The next business improvement is an "intervention" trial tied to "real outcomes". Additional documentation should include service complaints, tenure at each observation, plan changes, and previous retention contacts. Fairness checks are needed even when they can analyse the data, because it's a sensitive business consideration when senior staff are involved. A dashboard should show the current probability of churn, actual outcomes, model drift, and contact burden, without revealing too much personal information.

14. Overall Conclusion

The rebuilt churn project benefits from an enhanced technical context compared to a standalone ANN notebook: source verification, training-only clustering, model comparisons, validation-oriented decisions, and independently reported test scores. It states that churn varies significantly across the four published clusters and that the source dataset here has 6,741 records. Cluster 3 has the most numbers of risk and Cluster 1 is the least stable. These results do warrant further targeted research, and not a blanket incentive. The ANN classifies 91.90% of actual churners at the selected threshold, but it also produces a high number of false alerts. Logistic Regression can be shown to perform slightly better on several held-out metrics; it turns out that additional network complexity has not been proven to add business value. A limited, measured retention pilot, with both segmentation and individual risk scores is the next step suggested to be done. Measuring a customer's retention and net benefit incrementally should be the measure of success, not prediction scores.

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Appendix A: Data Dictionary and Encoding Notes

The following fields describe the encoded analytical inputs used for model training. Encoded columns should not be interpreted as raw customer-facing values. Tenure and monthly charges require the original scaling rules before live customer information can be scored.

Table 9. Predictors and operational meanings

Variable	Representation	Usage
gender, SeniorCitizen, Dependents	Binary numeric indicators	Customer characteristics
PhoneService, MultipleLines	Binary numeric indicators	Service selection
InternetService_DSL, InternetService_Fiber optic	Service flags	Internet product selection
Contract_Month-to-month, Contract_One year, Contract_Two year	One-hot indicators	Contract selection
tenure, MonthlyCharges	Standardised numerical inputs	Duration and monthly charge patterns
NewCluster_0 to NewCluster_3	Training-only K-Means indicators	Additional segment predictors
Churn	Binary target 0 or 1	Stayed or cancelled

Appendix B: Files Used

The analysis was prepared from cleaned_full_dataset.csv and clustered_customer_data.csv supplied by the project owner and linked to the Stage 2 repository. Supporting project artefacts include Customer_Churn_ANN_Colab.ipynb, churn_pipeline.py, predict.py, metrics.json, model_scores.csv, held_out_test_predictions.csv, the saved customer_churn_ann.keras model, preprocessing.joblib and generated graphics. The original published PCA and Cluster fields are retained for descriptive comparison but are excluded from supervised predictors.